The Logic Foundry – Cohort 2.0 Course Outline

Program: Frontend Web Development & AI Coding Tools
Duration: 4 Months Intensive Training + 2 Months Internship
Format: Fully Online
Website: academy.devphilips.com

Program Overview

The Logic Foundry Cohort 2.0 is a rigorous, project-driven bootcamp designed to take aspiring developers from absolute scratch to job-ready frontend engineers. Students will learn core web fundamentals, modern component-driven development with React, backend integration with Firebase, and leverage modern AI coding assistants to accelerate development workflows.

Month 1: Web Fundamentals & Modern CSS

- **Week 1: Introduction to the Web & HTML5**
    - How the web works, semantic HTML markup, forms, and accessibility best practices.
- **Week 2: CSS3 Styling & Layouts**
    - Box model, typography, color theory, and responsive design principles.
- **Week 3: Advanced Layouts with Flexbox & CSS Grid**
    - Building complex, adaptive multi-column layouts without frameworks.
- **Week 4: Tailwind CSS & Version Control (Git & GitHub)**
    - Utility-first CSS workflow, collaborative version control, branching, and pull requests.

Month 2: JavaScript Mastery (ES6+)

- **Week 5: JavaScript Basics & Control Flow**
    - Variables, data types, operators, conditionals, and loops.
- **Week 6: Functions, Scope & DOM Manipulation**
    - Function declarations vs expressions, arrow functions, selecting elements, and handling user events.
- **Week 7: Advanced JavaScript & Asynchronous Programming**

    - Arrays, objects, higher-order methods (`map`, `filter`, `reduce`), Promises, and `async/await`.
- **Week 8: Working with APIs & Local Storage**
    - Fetch API, HTTP requests, handling JSON data, error handling, and browser storage persistence.

Month 3: Modern Frontend with ReactJS & Firebase

- **Week 9: Introduction to React & Component Architecture**
    - Vite setup, JSX, component hierarchy, props, and passing data.
- **Week 10: React State & Hooks**
    - `useState`, `useEffect`, managing component lifecycle, and handling complex user interactions.
- **Week 11: Routing & Form Management**
    - React Router for multi-page applications, form validation, and controlled components.
- **Week 12: Backend-as-a-Service with Firebase**
    - Firebase Authentication (Email/Password & Google Auth) and Cloud Firestore (CRUD operations).

Month 4: AI Coding Tools, Capstone & Deployment

- **Week 13: Leveraging AI Coding Assistants**
    - Integrating AI tools (ChatGPT, GitHub Copilot, Cursor) into your daily workflow for debugging, refactoring, and rapid prototyping.
- **Week 14: Capstone Project Architecture**
    - Planning, structuring, and building a full-scale production-ready web application from scratch.
- **Week 15: Testing, Performance & Optimization**
    - Code optimization, Lighthouse audits, debugging tools, and responsive polishing.
- **Week 16: Deployment & Portfolio Building**
    - Deploying applications via Vercel, connecting custom domains, optimizing GitHub portfolios, and resume reviews.

Months 5 & 6: Real-World Internship

- **Hands-on Experience:** Placement into a 2-month structured internship program working on team-based production codebases, participating in code reviews, and building real client/product deliverables.